Data Structures - Quick Notes

Array

An array is a collection of elements stored at contiguous memory locations, identified by an index. Accessing an element by index takes O(1) time. Inserting or deleting an element in the middle takes O(n) time because subsequent elements must be shifted. Arrays have a fixed size once allocated, unless a dynamic array implementation is used.

Linked List

A linked list is a linear data structure where each element, called a node, contains data and a pointer to the next node. Unlike arrays, linked lists do not require contiguous memory and can grow or shrink dynamically. Accessing an element takes O(n) time since you must traverse from the head. Insertion and deletion at the head take O(1) time. Common types include singly linked list, doubly linked list, and circular linked list.

Stack

A stack is a linear data structure that follows the LIFO (Last In First Out) principle. Insertion and deletion happen only at one end, called the top. The main operations are push (insert an element), pop (remove the top element), and peek (view the top element without removing it). All operations run in O(1) time. Stacks are used in function call management, undo operations in editors, and expression evaluation such as converting infix to postfix.

Queue

A queue is a linear data structure that follows the FIFO (First In First Out) principle. Insertion happens at the rear and deletion happens at the front. The main operations are enqueue (insert an element) and dequeue (remove an element), both running in O(1) time with a proper implementation. Queues are used in task scheduling, handling requests in web servers, and breadth-first search (BFS) traversal of graphs and trees. A circular queue reuses empty slots at the front once elements are dequeued, avoiding wasted space.

Stack vs Queue

The key difference between a stack and a queue is the order in which elements are removed. A stack removes the most recently added element first (LIFO), while a queue removes the oldest added element first (FIFO). A stack only needs one pointer (top) to operate, while a queue typically needs two pointers (front and rear).

Binary Tree

A binary tree is a hierarchical data structure in which each node has at most two children, referred to as the left child and the right child. A binary search tree (BST) additionally requires that the left subtree of a node contains only values smaller than the node, and the right subtree contains only values greater. Searching, insertion, and deletion in a balanced BST take O(log n) time, but can degrade to O(n) in an unbalanced tree.

Graph

A graph is a non-linear data structure consisting of a set of vertices (nodes) connected by edges. Graphs can be directed or undirected, and weighted or unweighted. Common representations are the adjacency matrix, which uses O(V^2) space, and the adjacency list, which uses O(V+E) space and is preferred for sparse graphs. Graph traversal algorithms include breadth-first search (BFS), which uses a queue, and depth-first search (DFS), which uses a stack or recursion.

Sorting Algorithms

Bubble sort repeatedly swaps adjacent elements if they are in the wrong order, with O(n^2) time complexity in the worst case. Merge sort divides the array into halves, sorts each half recursively, and merges them, achieving O(n log n) time complexity in all cases. Quick sort picks a pivot element and partitions the array around it, with average time complexity O(n log n) but worst case O(n^2) when the pivot choices are poor.